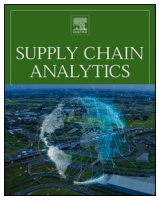
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A blockchain-based dynamic energy pricing model for supply chain resiliency using machine learning

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ABSTRACT

The escalation of energy prices and the pressing environmental concerns associated with excessive energy consumption have compelled consumers to adopt a more optimal approach towards energy usage and an advanced infrastructure such as smart grids. Blockchain technology significantly improves energy management by creating supply chain resiliency in a distributed smart grid. This study proposes a blockchain-based decision-making framework with a dynamic energy pricing model to manage energy distributions, particularly during an energy crisis. Empirical data from U.S. consumers are employed to show the applicability of the proposed model. We include price elasticity to address changes in energy market prices. Findings revealed that the proposed framework reduces total energy costs and performs better when a disruption has occurred. This study provides a post hoc analysis in which four machine learning algorithms are used to predict energy consumption. Results suggest that the autoregressive integrated moving average (ARIMA) algorithm has the highest accuracy compared to other algorithms.

1. Introduction

The escalation of energy prices and the pressing environmental concerns associated with excessive energy consumption have compelled consumers to adopt a more optimal approach towards energy usage and an advanced infrastructure such as smart grids. The 7th goal of sustainable development requires affordable and clean energy. However, according to US Energy Information Administration projections, energy consumption is expected to rise by 28% in 2040 [14]. Buildings and institutions account for nearly 36% of the world's total energy consumption [4]. The scarcity of energy resources has received considerable attention from individuals and organizations to optimize energy consumption. To ensure a sustainable future, balancing energy production and consumption is crucial [60]. Managing energy consumption is essential to achieving energy savings and analyzing the disparities between predicted and actual energy consumption, which can serve as a valuable decision-making tool [70].

Energy security is an essential requirement in improving economic growth and ensuring social welfare, which has resulted in a heavy reliance on reliable and resilient energy supply chains in modern and developing economies. Energy supply chain disruptions have considerable consequences, including economic and social costs, which makes energy management a primary global concern [52]. Severe weather events, such as winter storms and wildfires, can cause devastating power outages, posing significant challenges to community safety and welfare. For instance, the UK experienced power outages during the winter storms in February 2014, affecting over 80,000 households, which resulted in long-lasting disruptions [22]. Similarly, Texas faced power outages in 2021, resulting in the loss of life, widespread household disruptions, and significant economic losses in which around 4 million homes and businesses were left without power, along with 1000 deaths [23]. Other U.S. states also need energy supply chain resilience, e.g.,

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California had the most power outages in 2022. Thus, considerable importance has been placed on energy supply chain resilience.

Prior works have offered viable solutions to address supply chain resilience in energy management (e.g., [6,17] for a review of past studies). Previous studies suggested blockchain technology can contribute to energy management by restructuring energy information-sharing systems [27,6]. Blockchain technology enables secure, decentralized, and transparent record-keeping information systems for various industries, particularly the energy sector [26,51]. Using a distributed ledger information system, blockchain provides immutability and tamper-proof validation of all transactions within the network, which results in increasing trust and accountability among participants [41]. Blockchain offers visibility, which can improve supply chain performance during disruptions [55]. Blockchain excludes intermediaries in information-sharing systems, which create direct, peer-to-peer transactions [50], making blockchain an attractive solution for diverse sectors such as supply chain management [50,8], and energy [27]. Moreover, blockchain technology can improve energy supply chain efficiency by streamlining tasks and enhancing integration [19, 27]. With blockchain, complex energy supply chains can become more transparent, authentic, and trustworthy, offering a unified view of the entire process [55].

A significant body of literature explored the blockchain potential in energy supply chain resilience [3,6,7]. However, we must still contribute to the research gap by developing more analytical decision-making models to mitigate unbalanced energy supply demand, particularly during a natural disaster. Therefore, this paper addresses this research question: *How can blockchain contribute to energy consumption management during disruptions*? This paper addresses the research question by presenting a blockchain-based decision-making model. This paper presents a nonlinear programming problem to explain how peer-to-peer trading capabilities of blockchain technology can be included in energy distribution management. Moreover, this paper employs four machine learning algorithms to forecast energy demand by which demand variations can be addressed. Publicly available empirical data is used to show the applicability and validity of the proposed mathematical decision-making model. The effects of parameters are considered on the outcomes using the sensitivity analysis.

1.1. Contributions

This paper contributes to prior works by providing empirical evidence and real-world data to support the comparison between traditional and blockchain-based decision-making models. This empirical analysis can better understand the potential benefits and limitations of adopting blockchain technology for decision-making purposes. Moreover, this paper improves the decision-making process by integrating price elasticity to account for the price-demand effects on energy consumers' behavior. Price elasticity is important in explaining consumers' patterns in energy decision-making models. By considering price elasticity in decision-making models, we can explain how changes in energy prices influence consumers' choices and consumption behavior. To optimize energy consumption and profitability, this study proposes a novel blockchain-based framework that enables consumers to purchase energy at the most cost-effective rates while incentivizing energy conservation. Leveraging real-time pricing and penalties, the proposed system fine-tunes consumer demand in response to fluctuating energy prices, which results in minimizing consumption costs and maximizing profits. Blockchain technology ensures secure, transparent, and tamper-proof record-keeping, further reinforcing the efficacy of the proposed framework. By implementing a system of fines and rewards, the proposed approach sensitizes consumer behavior to the price and penalty of each energy unit, driving toward a more optimal and sustainable energy landscape. Moreover, this paper contributes to past work by employing four machine learning algorithms to predict energy consumption, which include regression, neural networks, K-nearest neighbor, and autoregressive integrated moving average (ARIMA) algorithm.

The rest of the paper is organized as follows. The next section provides a summary of prior work. Section 3 presents theoretical support for the proposed model. Section 4 proposes the mathematical model and presents the results. Section 5 discusses the findings and provides the conclusion.

2. Prior work

We explored the prior works available in Web of Science and Google Scholar databases using keywords of "blockchain," "resilience," "energy management," and "energy consumption."

2.1. Energy consumption management

Energy management refers to the strategic planning and utilization of energy production units and the control and monitoring of energy consumption, with the primary objectives of resource protection, climate mitigation, and cost savings. Energy consumers such as manufacturers must have continuous access to energy [9].

Yu et al. [68] proposed an effective policy framework to manage household energy demand by conducting customized surveys to gather essential data and developing a tailored policy to adjust household energy usage. Their findings showed that the existing policy creation process amongst various stakeholders is not optimal for achieving significant impacts from all policies. Moreover, Wang et al. [64] explored household power consumption management based on price and demand responses by designing a mathematical model to minimize energy consumption. They concluded that classifying loads is essential to reduce consumption, and most energy consumption comes from sheddable loads.

In transportation energy consumption management, González et al. [24] proposed optimizing energy consumption by analyzing the breakdown of urban rail energy usage, including a comprehensive review of genuine practices, plans, and equipment to decrease urban rail energy consumption. Their results showed that energy consumption in energy systems could be reduced by implementing timetables, driving plans, and performance control. In manufacturing companies' energy management, Introna et al. [30] introduced an organizational tool to reduce energy consumption in firms using general maturity model tools. Their approach manages energy demand and serves as an alternative to the certification of the organization's energy management system, consistent with the ISO 50001 standards. Moreover, Cai et al. [13] investigated the permissible amount of energy consumption of parts in the production industry. They developed a mathematical model to optimize energy consumption in industries. Their findings revealed that their proposed method was applicable for setting a workpiece's basic fine energy consumption allowance (FECA) and was reliable for setting the FECA of factories.

In a hospital's energy consumption management, Calcedo et al. [20] (evaluated the impact of maintenance management on energy saving. They designed a mathematical model to optimize energy and discovered that air conditioning, plumbing installations, and low-voltage systems consume more than 80% of the hospital's energy. Their solutions included preventive maintenance, which could result in a significant reduction in energy consumption.

Wang et al. [63] proposed a method for assessing energy consumption based on an energy use indicator in different parts of construction projects' energy management. Their evidence revealed that the sub-item energy consumption index derived from multi-functional structures exhibited lower error than single-functional structures, making it suitable for evaluating energy consumption in purely functional structures. Moreover, Lu et al. [42] developed an improved method to predict short-term building energy consumption by combining extreme gradient boosting (XGBoost) and complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN). Their proposed model yielded

lower errors than benchmark models, with satisfactory predictive functionality. From a quantitative perspective, Iqbal et al. [31] employed a zero-waste strategy to manage energy consumption using a nonlinear mathematical model for optimizing energy consumption. By reducing the energy required for producing main and secondary products and recycling packaging materials, they achieved an optimal planning horizon time.

Khan et al. [36] employed artificial intelligence techniques to predict and estimate power generation and consumption. Soni and Subhashini [58] discuss energy management's importance in addressing consumer demand. They proposed a smart grid with renewable energy integration with an energy management model for effective power consumption. The model includes load, temperature, weather, occupancy control, and synchronization between grid and green energy to reduce utility load and electricity costs.

Previous studies have addressed energy consumption management in various contexts, such as electric vehicles [59], home energy management [53], and construction [5]. However, we need more quantitative studies to explore the blockchain potential in energy consumption models.

2.2. Blockchain in energy management

Past empirical research shows that technology can considerably improve operational cost performance [45]. Blockchain technology helps consumers to control their energy management by offering a peer-to-peer trading capability [27], while many companies are lagging behind AI applications such as blockchain [54]. This innovation is primarily enabled through the technology's core feature: a distributed immutable ledger. This ledger offers a secure, tamper-proof system for recording and providing real-time updates on energy consumption data, ensuring data integrity and reliability [6]. According to Khatoon et al. [37], a wide variety of energy-related information, such as market values, edge costs (the costs associated with the delivery of electricity from the power plant to the consumer), legal regulations within the energy sector, and fuel prices, can be efficiently recorded and tracked. This capability allows for a comprehensive analysis of energy consumption patterns, offering insights into market trends.

Fig. 1 illustrates power systems that do not utilize blockchain technology, where energy consumers are not interconnected for peer-to-peer trading. In the disruption scenario depicted in Fig. 1, the central power generation unit faces disruption due to an adverse event, such as a severe storm, flood, or earthquake. This disruption leads to power distribution issues for consumers 3, 4, and 5. Conversely, Fig. 2 presents power systems that employ blockchain technology, enabling energy customers to be interconnected and trade energy through a blockchain-based platform. In this arrangement shown in Fig. 2, customers experiencing disruptions receive power from other consumers.

Burer et al. [12] examined the risks and opportunities associated with blockchain adoption in the energy sector. Their work offered a thorough understanding of blockchain technology's applications.

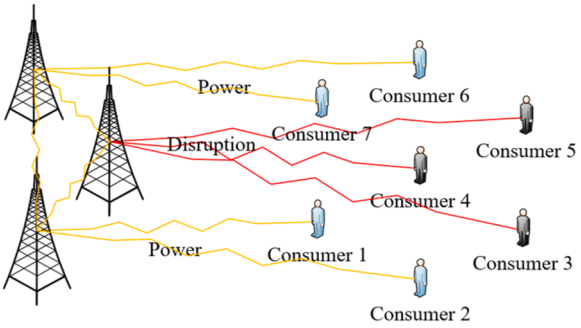


Fig. 1. Non-blockchain-based power systems.

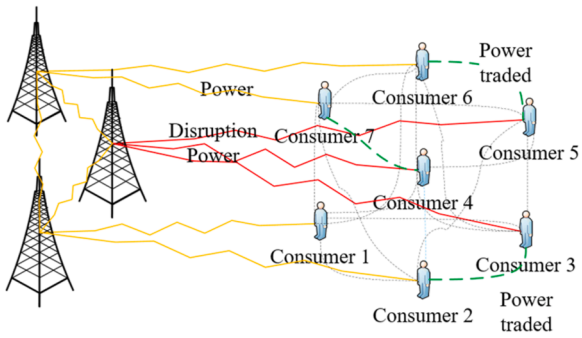


Fig. 2. Blockchain-based power systems.

Trading is an important factor in the energy market. Liu et al. [39] investigated the potential of peer-to-peer energy trading using distributed energy exchanges. Their findings demonstrated that matching electricity prices in real-time without intermediaries was feasible. Moreover, Wang et al. [63] extensively analyzed the challenges and limitations of blockchain-based energy trading. Their study also evaluated existing energy trading platforms. In addition, Foti and Vavalis [18] introduced a decentralized auction mechanism for energy trading. Their approach allowed real-time bidding at a fair price, meeting significant market demands.

Li et al. [38] presented a distributed energy system including heterogeneous renewable energy sources and end-consumers. Their results showed that monitoring power consumption across the system was achievable through party interactions, which can result in an efficient grid electricity purchasing program. Moreover, Hou et al. [28] evaluated the competitiveness of distributed energy systems. By implementing blockchain technology, they discovered that competition in distributed energy sources increased, along with several policy recommendations. For an improved secured platform, Khan et al. [36] proposed a blockchain-based framework, which includes multi-proof-of-work and proof-of-stack, to improve a smart grid and hierarchy of power distribution. A power generation company (PGC) can generate wholesale electricity in a distributed smart grid with reduced uncertainty and variability. Regarding consumer privacy and smart meter data, Zhao et al. [71] developed a blockchain framework based on privacy-preserving billing by which consumers could have ordinary bills for any period.

There are multiple review papers to explore the role of blockchain in energy management [7], particularly in the future of energy management [11], energy challenges and opportunities [6], the Internet of energy management [46]. Interested readers can follow Abed et al. [1] and Karumba et al. [35] to review the recent blockchain works in energy management. However, there is a lack of research to include blockchain technology for offering efficient and practical consumption patterns to customers in energy consumption management. In brief, Table 1 summarizes the contribution of this paper to prior works.

3. Methodology

We employed mathematical optimization to address the research question. We compare the outcomes (suggested consumptions) with current consumption to demonstrate the efficacy of the proposed model. This paper uses Python programming with Pyomo, which is an open-source optimization library. Note that the source code of Python and data are available here: https://github.com/russellpr/SCA2024. Finally, this paper performs the sensitivity analysis to evaluate the effects of parameters on the outcomes.

3.1. Data

We employed the data from U.S. grid systems to show the applica-

Table 1
Blockchain-based smart grid studies.

Study	Mathematical optimization	Consumption patterns	Price elasticity of demand	Empirical data	Machine learning
Hou et al. [29]	Yes	No	No	No	No
Zhou et al. [72]	No	No	No	Yes	Yes
Onyeka et al. (2020) [48]	Yes	No	No	Yes	No
Miglani et al. [46]	No	No	No	No	No
Wang et al. [65]	Yes	No	No	Yes	No
Jamil et al. [33]	Yes	Yes	No	Yes	Yes
Xu et al. [67]	Yes	Yes	No	No	No
Cui et al. [15]	Yes	No	No	Yes	No
J. Liu et al. [40]	No	No	No	Yes	Yes
Tsaousoglou et al. [61]	Yes	Yes	No	No	Yes
Zhang et al. [69]	Yes	No	No	Yes	No
X. Zhang et al. [69]	Yes	No	No	Yes	Yes
Xu et al. [66]	Yes	No	No	Yes	No
This paper	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

bility of the proposed model. The dataset encompasses power consumption data for North Carolina and South Carolina from June 2022 to July 2023. The source of data, which includes reginal energy consumptions, is the energy information administration (EIA), a U.S. governmental agency [16]. Table 2 provides details on the parameters. Here, the consumer's base demand (b_i) is calculated as their minimum consumption (d_i) average. The tariff is half the energy price, and the maximum energy consumption is $1.5d_i$. Moreover, we estimate the price elasticity based on price and demand in Table 2.

3.2. The proposed model

We consider a grid system where all users (e.g., companies) have blockchain-based smart meters. There are defined smart contracts regarding energy prices and limitations. There are peer-to-peer trading capabilities in the system. It is assumed that energy prices in the market and among users are higher than energy providers. Table 3 shows notations used in the mathematical model.

The mathematical model is,

$$\text{Min } Z = \sum_i D_i\, p_i + \sum_i t_i(D_i - b_i) \tag{1}$$

Subject to,

$$D_i \geq d_i/2 - e_i(p_i + t_i); \forall i \in I \tag{2}$$

$$D_i \leq c_i; \forall i \in I \tag{3}$$

$$e_i = \frac{(d_{i-1} - d_i)}{(p_{i-1} + t_{i-1}) - (p_i + t_i)} \times \frac{(p_{i-1} + t_{i-1}) + (p_i + t_i)}{(d_{i-1} + d_i)}; \forall i \in I \tag{4}$$

$$D_i > 0; \forall i \in I \tag{5}$$

Table 3
Notations.

i	Index for time, $i \in \{1, 2, 3, \ldots, I\}$
b_i	Consumer's base demand (average of minimum consumption) for energy at the time of i
c_i	Maximum consumer's energy consumption at the time of i,
d_i	Consumer's consumption for energy at the time of i
e_i	The coefficients of price elasticity of the demand at the time of i
p_i	Energy price at the time of i
t_i	Tariff at time of i (reward if $D_i < b_i$ or penalty if $D_i > b_i$)
D_i	Suggested consumer demand (from energy providers) for energy at the time of i

where, $t_i(D_i - b_i) > 0$ indicates a cost for the consumer, while $t_i(D_i - b_i) < 0$ indicates benefits for the consumer. Therefore, Eq. (1) minimizes the total cost of each consumer in one day including tariff rate [21,57]. Eq. (2) and Eq. (3) control the suggested demand within its bounds. Eq. (4) shows the price elasticity of the demand at time of i [49]. Eq. (5) shows the variable domain.

4. Result

In this section, we evaluate the applicability of the proposed model using data from North and South Carolina. Table 4 and Fig. 3 show the results, including the actual (d_i) and suggested demands (D_i). The findings reveal that the proposed model significantly improves consumption (t-value = 12.08, p-value < 0.001, $df = 26$). Our model proposes a significant reduction in electricity consumption. This significant decrease yields substantial savings for businesses and fosters energy conservation. The optimized demand facilitates more efficient operation of power plants, resulting in decreased greenhouse gas emissions. Moreover, maintaining a balanced grid load diminishes the risk of

Table 2
Parameters.

Month_i	d_i (mWh)	p_i (cent)/(mWh)	b_i	t_i	c_i	e_i
Jun-22	20,701,722	0.00981	18,713,502	0.0049	31,052,583	0.954
Jul-22	22,860,739	0.01018	18,713,502	0.0051	34,291,109	2.677
Aug-22	21,805,571	0.01053	18,713,502	0.0053	32,708,357	-1.397
Sep-22	18,281,823	0.01079	18,713,502	0.0054	27,422,735	-7.207
Oct-22	15,892,131	0.01052	18,713,502	0.0053	23,838,197	5.519
Nov-22	16,666,861	0.00982	18,713,502	0.0049	25,000,292	-0.691
Dec-22	20,489,687	0.01040	18,713,502	0.0052	30,734,531	3.586
Jan-23	18,795,106	0.01092	18,713,502	0.0055	28,192,659	-1.768
Feb-23	16,027,793	0.01080	18,713,502	0.0054	24,041,690	14.383
Mar-23	16,953,225	0.01039	18,713,502	0.0052	25,429,838	-1.450
Apr-23	15,643,207	0.01036	18,713,502	0.0052	23,464,811	27.790
May-23	16,612,600	0.01022	18,713,502	0.0051	24,918,900	-4.418
Jun-23	18,447,338	0.01035	18,713,502	0.0052	27,671,007	8.280
Jul-23	22,811,228	0.01086	18,713,502	0.0054	34,216,842	4.399

Table 4
Comparisons Results.

$Month_i$	Consumer's base demand (d_i)	Suggested consumer demand (D_i)
Jun-22	20,701,722	10,350,860
Jul-22	22,860,739	11,430,369
Aug-22	21,805,571	10,902,785
Sep-22	18,281,823	9,140,911
Oct-22	15,892,131	7,946,065
Nov-22	16,666,861	8,333,430
Dec-22	20,489,687	10,244,843
Jan-23	18,795,106	9,397,552
Feb-23	16,027,793	8,013,896
Mar-23	16,953,225	8,476,612
Apr-23	15,643,207	7,821,602
May-23	16,612,600	8,306,299
Jun-23	18,447,338	9,223,668
Jul-23	22,811,228	11,405,613

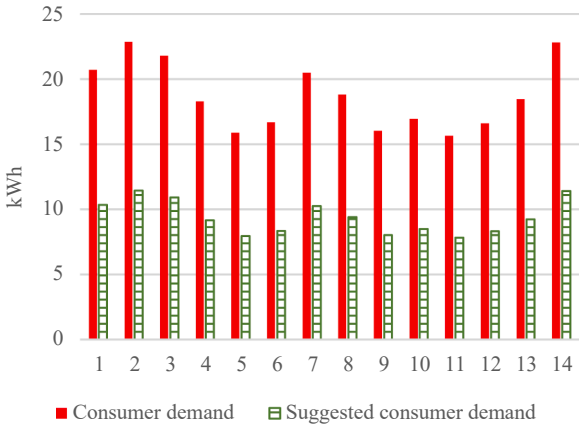


Fig. 3. Comparisons Results in 10*-6.

blackouts and unnecessary energy wastage. Implementing this model paves the way for a more sustainable and cost-effective electricity consumption paradigm within the industry.

4.1. Sensitivity analysis

To evaluate the effects of inputs on the outputs, this paper performed a sensitivity analysis test for two parameters: consumer's base demand (d_i) and price (p_i). Each parameter increases and decreases by 10%, 20%, and 30%. The outcomes are shown in Tables 5 and 6. As demand increases, the total cost of each consumer is expected to increase, as shown in Table 5. This correlation highlights the vital role of demand in determining the outcomes of the total cost of each consumer. Such an observation is crucial for businesses or stakeholders to predict and strategize operations in scenarios where market demand fluctuates.

Table 6 illustrates that as the price escalates, the total cost incurred by each consumer also rises. Increasing electricity prices can incentivize consumers to reduce their consumption to economize. However, certain essential appliances, such as machines and computer systems, necessitate continuous power, limiting the potential for significant consumption reduction. In our research, we investigated the repercussions of fluctuations in electricity prices. Even with a marginal decrease in consumption, an elevated price inevitably results in a higher aggregate expenditure for consumers.

Table 5
Effect of demand on the outcomes.

d_i	Objective function
-30%	478,166
-20%	546,476
-10%	614,785
0%	683,095
10%	751,404
20%	819,714
30%	888,023

Table 6
Effect of price on the outcomes.

p_i	Objective function
-30%	478,166
-20%	546,476
-10%	614,785
0%	683,095
10%	751,404
20%	819,714
30%	888,023

4.2. Post Hoc analysis

Machine learning algorithms can handle complex, non-linear relationships in data, adapt to changing patterns, and provide accurate forecasts [2,32,47]. Energy demand is influenced by various factors, including weather conditions, economic activities, consumer behavior, and technological changes [62]. Machine learning algorithms can capture these complex relationships and interactions, something traditional statistical methods might struggle with [10]. With the advent of smart grids and the Internet of Things (IoT) devices, there's a surge in real-time data. Machine learning can process and analyze this voluminous data effectively, providing timely and accurate forecasts for real-time decision-making [10]. Machine learning models, especially those employing advanced techniques like neural networks or ensemble methods, often outperform traditional models in terms of predictive accuracy [44]. Machine learning models can adapt to changing patterns in energy usage. This is increasingly important in a world where energy consumption patterns evolve due to factors like renewable energy sources and electric vehicles. Accurate demand forecasts help optimize energy production, distribution, and storage, reducing wastage and costs. With the growing emphasis on renewable energy sources, which are inherently variable, accurate demand forecasting becomes even more critical to balance the grid effectively.

This paper employed machine learning algorithms to predict energy consumption. We compare the algorithms to propose a better algorithm to forecast the demand, which is a critical factor in decision-making processes in energy management. The algorithms are regression, neural networks, K-nearest neighbor, and ARIMA algorithm. The ARIMA algorithm excels at identifying trends and seasonal patterns in data, which makes it effective for short-term forecasting. Moreover, ARIMA's integrated component aids in stabilizing the data, a necessary step to uncover significant statistical characteristics. We use a trial-and-error approach to tune the parameter for the machine learning algorithms regarding the lowest root-mean-square deviation (RMSE) index. This approach, also known as manual tuning, involves iteratively adjusting the parameters, running the model, and observing the performance to find the best values for parameters. We relied on actual model outputs to make decisions about parameter adjustments. Our goal was to balance generalization with model specificity. We aimed to avoid a one-size-fits-all approach, recognizing that the optimal parameters for one dataset or problem might not be the best for another.

4.2.1. Regression

Regression analysis facilitates understanding how dependent variables change in response to alterations in independent variables, making it indispensable for predicting trends and making informed decisions. The efficacy of regression analysis is contingent upon the correct specification of the model and the assumptions underlying it. Misspecification can lead to biased and unreliable forecasts. Furthermore,

unless specifically designed, regression models are inherently limited in capturing non-linear relationships. This limitation is critical in complex systems where interactions are not strictly linear. Moreover, regression analysis often requires large datasets to produce accurate predictions and is vulnerable to the influence of outliers, which can skew results significantly. Regression assumptions are met in this paper (e.g., multicollinearity, normality, and homoscedasticity).

4.2.2. Neural networks

Neural networks can process multiple inputs simultaneously, benefiting from parallel processing capabilities. Neural networks handle complex, nonlinear data processing and offer efficient learning and prediction, making them suitable for predictive analytics in electricity demand forecasting [56]. Neural networks often require a large amount of data to train effectively, which can be a limitation when data is scarce or expensive. The model was compiled using the adaptive moment estimation optimizer and mean squared error loss function, which are well-suited for regression problems. The model consists of a sequential architecture with three layers. The neural network's structure was designed with 10, 8, and 1 neuron in each layer. The first two layers employ rectified linear unit (ReLU) activation functions, optimizing the model's ability to capture nonlinear relationships in the data. The final layer uses a linear activation function, fitting for the regression-based electricity demand prediction. We trained the model using 80% of the data and tested it on the remaining 20%.

4.2.3. K-nearest neighbor

K-nearest neighbors (KNN) is a powerful non-parametric algorithm that has shown considerable ability in handling time-series data [25]. One of the main key benefits of KNN is the absence of a training phase, as it uses the whole dataset for 'training' during prediction, enabling rapid model development. However, features in KNN need to be normalized. Choosing the right value of K (here is 5) is crucial for the algorithm's performance. Also, KNN can perform poorly in scenarios with high-dimensional data.

4.2.4. ARIMA

ARIMA algorithm is ideal for forecasting time series data like electricity demand [34]. One of the ARIMA algorithm's primary advantages is capturing the trend and seasonality in data, making it suitable for short-term forecasting where these elements are prevalent. ARIMA's integrated component helps make the data stationary, a prerequisite for revealing meaningful statistical properties. Moreover, the model's parameters can be systematically identified and estimated based on autocorrelation and partial autocorrelation functions. However, ARIMA also has notable disadvantages. Its effectiveness is limited to linear relationships, making it less suitable for data with complex, non-linear patterns. The model also assumes a constant variance over time (homoscedasticity), which might not hold for all time series. Moreover, ARIMA requires the series to be stationary or transformed to be stationary, which can be a complex process for certain data sets. It also struggles with handling missing values and requires extensive historical data for accurate forecasting. Our model was configured with parameters (p=1, d=1, q=1) representing the autoregressive, integrated, and moving average components. This configuration was chosen after thorough analysis, including examining autocorrelation and partial autocorrelation plots.

4.2.5. Comparison results

We implemented the algorithms in a Python environment. Inputs are consumption time (year, month, day, and hour). Outputs herein are energy consumption. Table 7 shows the comparison results. The findings show that the ARIMA algorithm predicts energy consumption better than others. ARIMA has the highest R^2, which explains the variation of outcomes. Moreover, ARIMA has the lowest root-mean-square deviation (RMSE), indicating algorithm accuracy errors.

Table 7
Machine learning algorithms comparisons.

Algorithm	R^2	RMSE
Regression	0.18	4950.64
Neural Network	0.18	4962.82
KNN	0.96	1059.97
ARIMA	0.98	558.48

5. Discussion and conclusion

The challenge facing incorporating renewable energy sources into the grid stems from their inherently unpredictable nature, closely tied to weather conditions. Renewable energy sources significantly are related to weather conditions. For example, in February 2014, winter storms in the UK led to power outages impacting more than 80,000 homes, causing prolonged disturbances [22]. In a similar vein, Texas suffered power failures in 2021, leading to fatalities, extensive household disturbances, and considerable economic damage [23]. Approximately 4 million homes and businesses were deprived of electricity, and there were about 1000 fatalities [23]. To enhance the reliability of these intermittent energy sources, adaptive and responsive measures must be taken, such as integrating optimized supply and demand responses within the energy system [43]. As a means of addressing the uncertainty associated with distributed energy sources and systems, numerous decision-makers have embraced digital technologies, deploying smart meters widely to exercise control over the energy infrastructure [73].

Various scholarly endeavors have underscored blockchain's role in modernizing energy management systems. This paper contributes to this growing field by offering a novel decision-making framework rooted in blockchain. Our proposed framework, grounded in the dynamic energy pricing model, promises substantial improvements in energy distribution, especially during energy crises. The proposed framework significantly contributes to sustainable development goals by providing affordable energy.

Empirical data from North and South Carolina have been vital in demonstrating the efficacy of our model. A significant reduction in electricity consumption, as observed in Table 4, resonates with the core objectives of sustainable energy management. By curtailing consumption, businesses are poised to realize substantial economic benefits. Moreover, the environmental implications of reduced greenhouse gas emissions cannot be understated. In the backdrop of global warming concerns, optimized energy consumption facilitated by our model could be a harbinger of sustainable energy management practices.

Our sensitivity analysis further elucidated the intricate relationship between consumer demand and price and overall energy costs. Findings from Tables 4 and 5 depict a predictable yet crucial correlation between these parameters. Understanding these dynamics becomes indispensable for businesses and stakeholders as the energy sector grapples with market volatility. They are empowered to make informed decisions, especially in an environment characterized by fluctuating market demands or price adjustments.

Furthermore, the realm of artificial intelligence and machine learning has also been explored in our study, specifically focusing on energy consumption prediction. The juxtaposition of various algorithms, as evident in Table 6, yielded intriguing insights. While all algorithms provided meaningful predictions, the ARIMA algorithm stood out regarding accuracy and reliability. This underlines its potential role as a tool for informed decision-making in energy management. While ARIMA models are powerful for time series forecasting, they can present several challenges that users must navigate. For example, ARIMA models necessitate that the time series data be stationary, meaning that its statistical properties such as mean, variance, and autocorrelation are constant over time. Achieving stationarity often requires differencing the data, which can complicate the analysis and interpretation. Moreover, the flexibility of ARIMA models, especially with many parameters,

raises the risk of overfitting. Overfitting occurs when the model captures noise rather than the underlying trend, leading to poor predictive performance on new data. Understanding ARIMA barriers is crucial for effectively applying ARIMA modeling in practical settings.

5.1. Managerial implications

The findings of this study offer several implications for managers and decision-makers within the energy sector. First, the demonstrated effectiveness of a blockchain-based decision-making framework in improving energy distribution suggests a substantial advantage in investing in blockchain technology. Particularly, the proposed model is a practical tool if sustainable development goals are focused. Managers should consider dedicating resources to further explore and implement blockchain solutions to stay competitive and responsive to energy crises. Second, the balanced grid load resulting from the proposed model can reduce the risk of blackouts and unnecessary energy wastage, implying a more reliable energy supply vital for a continuous power supply. Managers in sectors such as manufacturing and information technology (IT) can particularly benefit from such stability. Last but not least, the superior performance of the ARIMA algorithm in predicting energy consumption underlines the importance of integrating advanced analytical tools into energy management. Managers should consider upskilling their teams or collaborating with experts to harness the potential of machine learning in forecasting and planning.

5.2. Theoretical implications and future research

The results from this study underscore blockchain's potential to revolutionize energy management. This calls for re-examining existing theoretical frameworks on energy supply chain resilience, introducing blockchain as a core component. Moreover, incorporating a dynamic energy pricing model in a blockchain framework is an innovative cross-disciplinary approach. In addition, the successful employment of machine learning algorithms, particularly the ARIMA algorithm, to predict energy consumption presents a theoretical convergence of data science and energy management. This signals the need for a renewed exploration into how modern analytical tools can be woven into energy theories.

While this study focused on energy management, future research could delve into other blockchain applications in the energy sector, such as grid management or renewable energy certification. Moreover, a comprehensive study on dynamic pricing, considering factors like peak demand hours, seasonal changes, or renewable energy input, would provide richer insights and might result in refined pricing models. Future research could explore how integrating blockchain solutions with the IoT devices in smart homes or industrial setups can further optimize energy consumption. Determining the appropriate order of ARIMA models (i.e., the values of p, d, and q) requires a deep understanding of the underlying data and significant trial and error. Selecting incorrect parameters can lead to poor model performance. Future work can tune the parameter of the machine-learning algorithms to ensure the high quality of outcomes.

5.3. Limitations

The primary data utilized in this study is centered around North and South Carolina, potentially limiting the generalizability of the findings on a broader scale. These states' energy consumption patterns, regulations, and infrastructure might not mirror those in other regions or countries. Extending the research to encompass different regions or countries with varied energy landscapes would offer a more holistic understanding and validate the universality of the findings. While the ARIMA algorithm was the most effective in predicting energy consumption among the tested, it's worth noting that the machine learning field is rapidly evolving. The algorithms may not encompass the full spectrum of available or upcoming tools. Finally, factors such as governmental regulations, cultural attitudes toward energy conservation, and advancements in renewable energy sources, which can significantly impact energy consumption and pricing, were not the primary focus of this study.

CRediT authorship contribution statement

Moein: Writing – original draft, The Proposed Model, Methodology, Visualization. **Russell:** Writing – original draft, Machine learning, Methodology, Data curation, Optimization, Visualization. **Ava:** Writing – review & editing, Machine learning, Conceptualization, Supervision. **Omid:** Validation, Conceptualization. **Amirehsan:** Methodology, Investigation.

Declaration of Competing Interest

Authors have no conflict of interests to declare.

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